

CSDS 302 Discrete Math | Fall 2026 | Homework 5

Instructions

Section 5.1: 6, 60

Section 5.2: 3, 7, 12, 29

Section 5.3: 4, 7, 15

Section 5.4: 10, 13, 16, 17

Feedback

Score: 100%

Good work.

HW 5

5.1: 6, 60

⑥ $f_n = \lfloor \frac{n}{4} \rfloor \cdot 4, n \geq 1$

$$f_1 = \lfloor \frac{1}{4} \rfloor \cdot 4 = 0$$

$$f_2 = \lfloor \frac{2}{4} \rfloor \cdot 4 = 0$$

$$f_3 = \lfloor \frac{3}{4} \rfloor \cdot 4 = 0$$

$$f_4 = \lfloor 1 \rfloor \cdot 4 = 4$$

⑥⑦ $2 \cdot \sum_{k=1}^n (3k^2 + 4) + 5 \cdot \sum_{k=1}^n (2k^2 - 1)$

$$\sum_{k=1}^n 2(3k^2 + 4) + 5(2k^2 - 1)$$

$$= \sum_{k=1}^n 6k^2 + 8 + 10k^2 - 5$$

$$= \sum_{k=1}^n 16k^2 + 3$$

5.2 3, 7, 12, 29

③ $P(n) = \frac{n(n+1)(2n+1)}{6}$

a $P(1) = \frac{1(2)(3)}{6} = 1 = 1^2 = \text{true}$

b $P(k) = \frac{k(k+1)(2k+1)}{6} = 1^2 + 2^2 + \dots + k^2$

c, d $P(k+1) = \frac{(k+1)(k+2)(2k+3)}{6} = 1^2 + 2^2 + \dots + k^2 + (k+1)^2$

From the previous step, we know $1^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$

$P(k+1) = \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} = \text{RHS}$

$\frac{k(k+1)(2k+1)}{6} = \frac{(k^2+k)(2k+1)}{6} = \text{RHS}^*$

$\frac{2k^3 + 6k^2 + 4k + 3k^2 + 9k + 6}{6} =$

$= \frac{2k^3 + 9k^2 + 13k + 6}{6}$

$= \frac{2k^3 + 9k^2 + 13k + 6}{6} = \text{RHS}^*$

TRUE

7) $n \geq 1$
 $1 + 6 + 11 + 16 + \dots + (5n-4) = \frac{n(5n-3)}{2}$

$P(1): 1 = \frac{1(5-3)}{2}$ TRUE

$P(k): 1 + 6 + \dots + (5k-4) = \frac{k(5k-3)}{2}$

Assume $P(k)$ is true

$P(k+1): 1 + 6 + \dots + (5k-4) + (5k+1) = \frac{(k+1)(5k+2)}{2}$
 $\hookrightarrow \frac{k(5k-3)}{2}$ by sup $P(k)$

$\frac{k(5k-3)}{2} + (5k+1) = \frac{(k+1)(5k+2)}{2}$ *

$\frac{5k^2 + 7k + 2}{2} = \frac{(k+1)(5k+2)}{2}$ *

$\frac{(5k+2)(k+1)}{2} = \frac{(k+1)(5k+2)}{2}$ *

TRUE BY M.I.

12) $\frac{1}{2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}; n \geq 1$

$P(1) = \frac{1}{2} = \frac{1}{1+1}$ TRUE

$P(k) = \frac{1}{2} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$

$P(k+1): \frac{1}{2} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$ *
 $\frac{k}{k+1}$ (from $P(k)$)

$\frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$ *

$\frac{k^2 + 2k + 1}{(k+1)(k+2)} = \frac{k+1}{k+2}$ *

$\frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2}$ *

TRUE BY M.I.

29) $1 - 2 + 2^2 - 2^3 + \dots + (-1)^n 2^n, n \in \mathbb{Z}^+$

$\sum_{i=1}^n r^i = \frac{r^{n+1} - 1}{r - 1} \quad (-2)^n$

$\sum_{i=1}^n (-1)^i (2^i) = \frac{(-1)^{n+1} (2)^{n+1} - 1}{(-1)(2) - 1} = \frac{(-2)^{n+1} - 1}{-3}$

5.3 4, 7, 15

4) $5^n - 1$ is divisible by 4

a) $P(0): 4 \mid 5^0 - 1$
 $4 \mid 4$ TRUE

b) $P(k): 5^k - 1$ is divisible by 4

c) $P(k+1): 5^{k+1} - 1$

$\forall k$ for $k \geq 0$

if $5^k - 1$ is divisible by 4, then $5^{k+1} - 1$ is div by 4

7) $n, P(n)$ T_i beats T_{i+1} $i=1, 2, \dots, n$

a) $P(2) = T_2$ beats T_3 true

b) $P(k) = T_k$ beats $T_{(k+1)}$

c) $P(k+1) = T_{k+1}$ beats $T_{(k+2)}$

d) if T_k beats T_{k+1} then T_{k+1} beats T_{k+2}

15) $n(n^2+5)$ is divisible by 6 for $n \geq 0$
 $P(0) = 0$ div by 6 $\Rightarrow$ true
 $P(1) = 1(5+1) = 6$ div 6 $\Rightarrow$ TRUE

Hypothesis

$P(i)$ where $0 \leq i \leq n$ is true

Let's $P(n+1)$ is true

$$P(n) = n(n^2+5) \Rightarrow \text{TRUE} = n^3 + 5n$$

$$P(n+1) = (n+1)(n^2+2n+6) = n^3 + 3n^2 + 8n + 6$$

$$P(n+1) = \underbrace{(n^3 + 5n)}_{P(n)} + 3n^2 + 3n + 6$$

$P(n) \Rightarrow$ divisible by 6

Case 1: $n = 2k$ (n is even)

$$3(4k^2 + 2k + 2) \Rightarrow 6(2k^2 + k + 1)$$

$\in \mathbb{Z}$ since a sum of integers is an integer

Thus when n is even, $P(k+1)$ is true or div by 6

Case 2: $n = 2k+1$ (n is odd)

$$3(4k^2 + 6k + 2) \Rightarrow 6(2k^2 + k + 1)$$

$$6(2k^2 + k + 1)$$

$\in \mathbb{Z}$ since a sum of integers is an integer

Thus when n is odd $P(k+1)$ is div by 6

5.4 10, 13, 16, 17

10) $n = 3a + 5b$ where $n, a, b \in \mathbb{Z}$
for $n \geq 8$. $P(8) = 3 \cdot 1 + 5 \cdot 1$ true
 $P(9) = 3 \cdot 3 + 5 \cdot 0$ true

assume: for every i , $8 \leq i \leq n$, $P(i)$ is true

$P(n) = 3a + 5b$ is true

$$P(n+1) = P(n) + 1 = 3(a+2) + 5(b-1)$$

$$3a + 6 + 5b - 5 = 3a + 5b + 1$$

Therefore it's True

13) $P(2)$ is true since 2 is a prime number

Hypothesis: for $\forall 1 < k \leq n$,

$P(k) = ab$ where $1 < a, b < n$

$$a = p_1 p_2 \dots p_a \quad b = p_1 p_2 p_b$$

$$P(ab) = p_1 p_2 p_3 p_a p_b$$

$P(k)$ is true therefore $P(k+1)$ is also true

16) $\forall n, n \geq 2$

$P(2)$: sum of 2 odd integers is even:

$$(2k+1) + (2m+1) = 2(k+m+1) \Rightarrow \text{TRUE}$$

$P(i)$ where $2 \leq i \leq k$ is true then $P(n+1)$ is true

Let $k \geq 2$

Let O_1, O_2, O_3 be odd integers

Case 1: $n = k+1$ even

Then $O_1 + O_2 + O_3 + \dots + O_k + O_{k+1}$ then

odd & k is odd because $k+1$ is even.

So then $k+1$ odd integers

$$(O_1 + O_2 + \dots + O_k) + O_{k+1} \Rightarrow \text{is even}$$

Therefore $P(n)$ when n is even is true

Case 2: $n = k+1$ odd

Then $O_1 + O_2 + O_3 + \dots + O_k + O_{k+1}$ then

is even & k is even since $k+1$ is odd so

then $k+1$ odd integers

$$(O_1 + O_2 + \dots + O_k) + O_{k+1} \Rightarrow \text{is odd}$$

Therefore $P(n)$ when n is odd is true

17) $4^1 = 4$ $4^4 = 256$ $4^7 = 16384$
 $4^2 = 16$ $4^5 = 1024$ $4^8 = 65536$
 $4^3 = 64$ $4^6 = 4096$

The last digit of 4^n is 4 if n is odd
and is 6 if n is even

$P(1)$ is true $4 \Rightarrow 4$

$P(2)$ is true $16 \Rightarrow 6$

Let $k \geq 2$, $P(i)$ $1 \leq i \leq k$ true then

$P(k+1)$ is true.

Case 1: $(k+1)$ is even

Then k is odd, then the last digit of 4^k is

4 ; $4^k = 10m + 4, m \in \mathbb{Z}$
 $4^{k+1} = 4^k \cdot 4 = 4(10m + 4) = 10(4m + 4) + 4$

Therefore the last digit when n is even is 6.

Case 2: $(k+1)$ is odd

Then k is even, then the last digit of 4^k should be 6

$4^k = 10m + 6, m \in \mathbb{Z}^+$

$4^{k+1} = 4(10m + 6) = 10(4m + 2) + 4$

Therefore the last digit when n is odd is 4.